

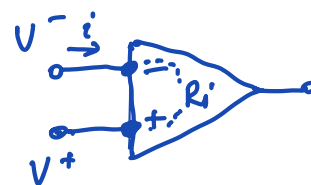
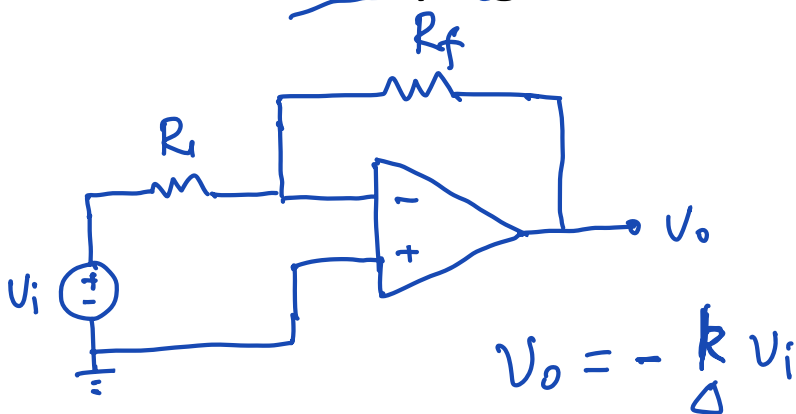
L#19: Inverting and non-inverting amplifier

1. Configuration of inverting amplifier

- Signal inputs from inverting terminal;
- Has a feedback loop;
- Non-inverting terminal is grounded.

2. Function

- Reverse the polarity of input signal while amplifying it.

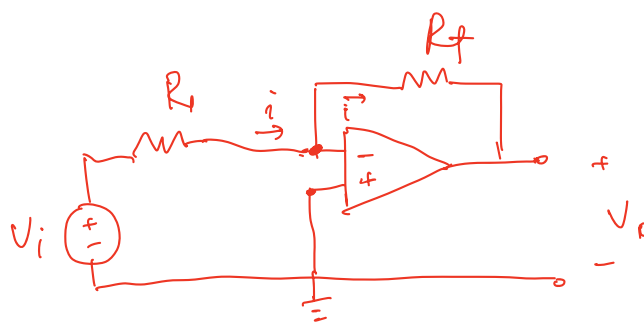


$$A \rightarrow > 10^5$$
$$R_i \rightarrow \infty$$

$$V^+ = V^-$$

$$i \rightarrow 0$$

$$R_i \rightarrow \infty$$



$$\frac{V_i - 0}{R_i} = \frac{0 - V_o}{R_f}$$

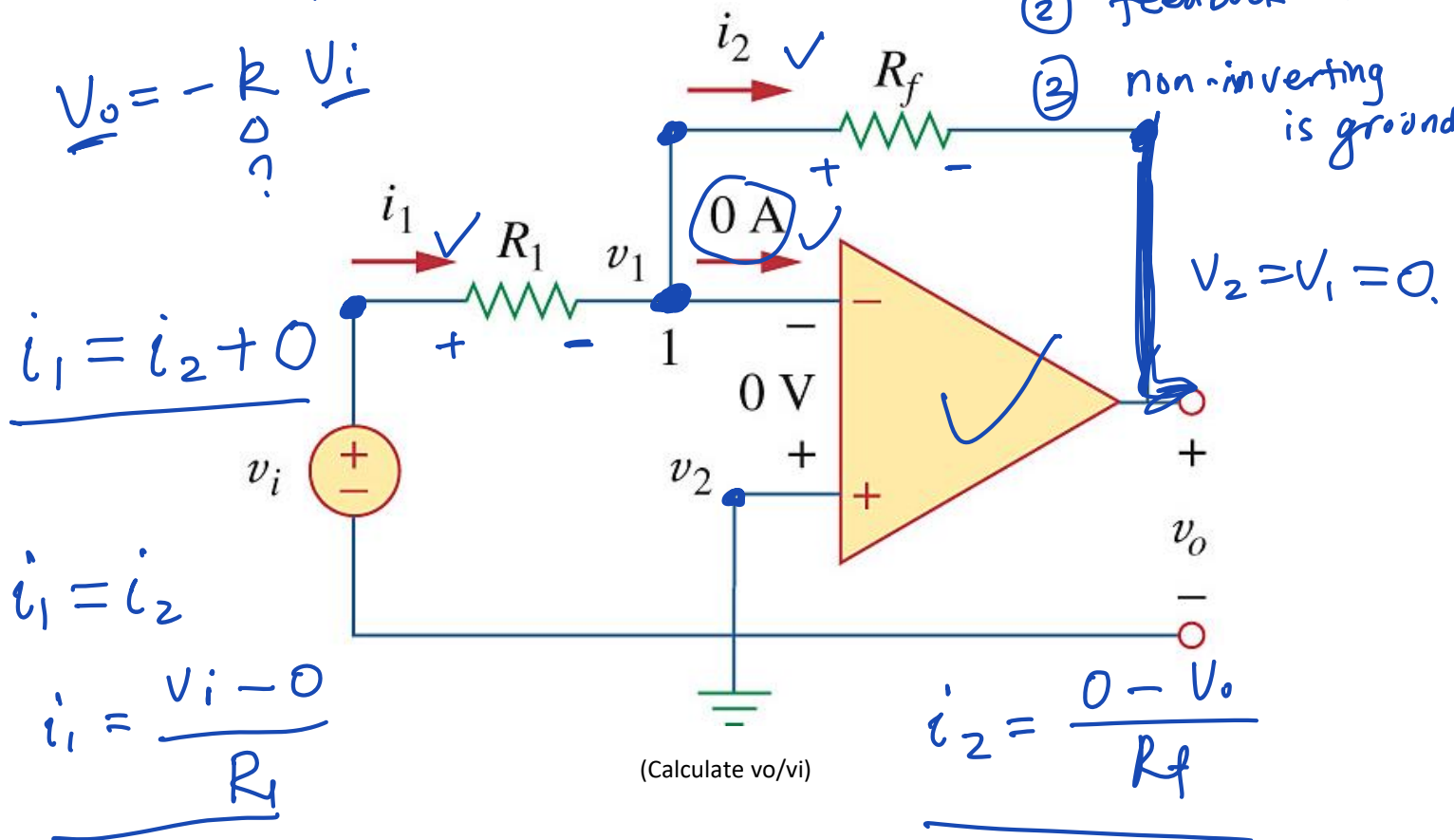
$$\Rightarrow V_o = -\frac{R_f}{R_i} V_i$$

Inverting amplifier

① input → inverting

② feedback ✓.

③ non-inverting is grounded



$$\frac{V_i}{R_1} = \frac{-V_o}{R_f} \Rightarrow V_o = -\left[\frac{R_f}{R_1}\right] \cdot V_i$$

$$V_i = 1.89 \text{ V}$$

$$\frac{R_f}{R_1} = 299$$

$$V_o = -1.89 \cdot 299$$

↑

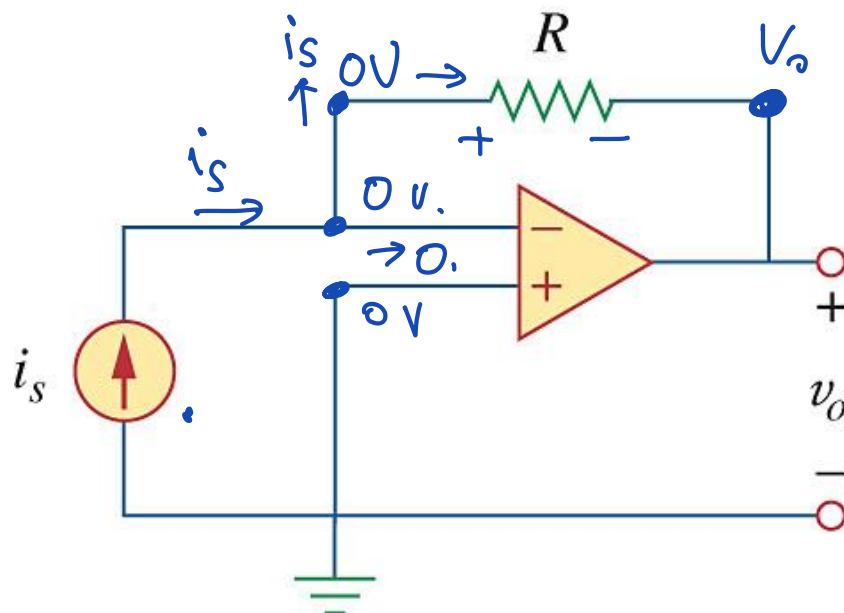
$$R_f = 3 \text{ k}\Omega$$

$$R_1 = 1 \text{ k}\Omega$$

$$\frac{R_f}{R_1} = 3$$

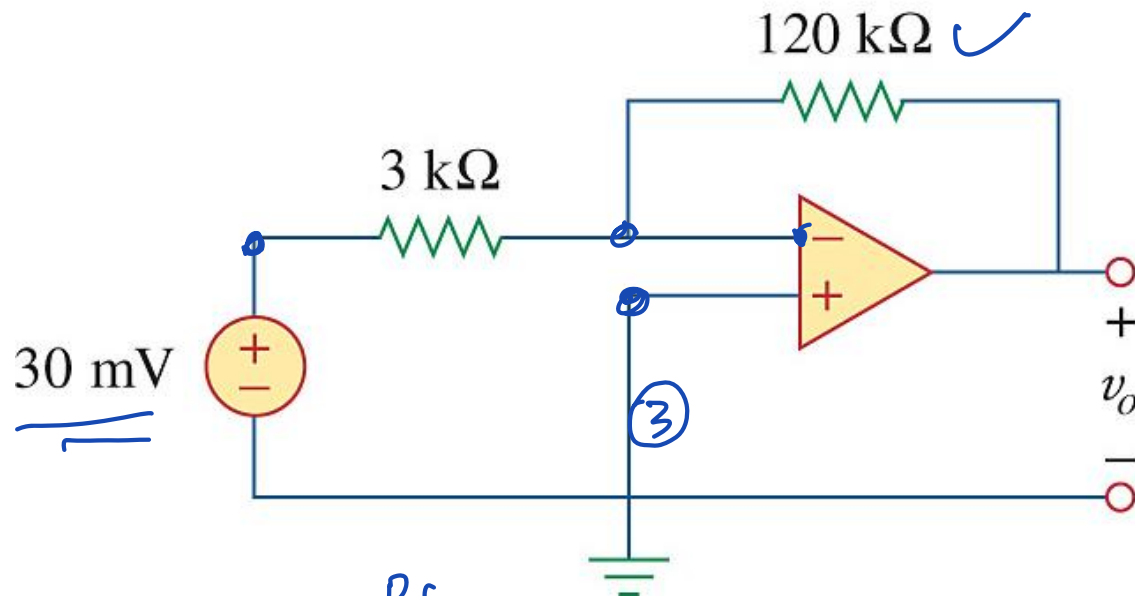
$$V_o = -3 V_i$$

Current to voltage converter



$$i_s = \frac{0 - V_o}{R} \Rightarrow V_o = \underline{\underline{-i_s R}}$$

Example #1: Determine the output voltage and current through the feedback resistor.



$$V_o = - \frac{R_f}{R_i} 30 \times 10^{-3} \text{ V.}$$

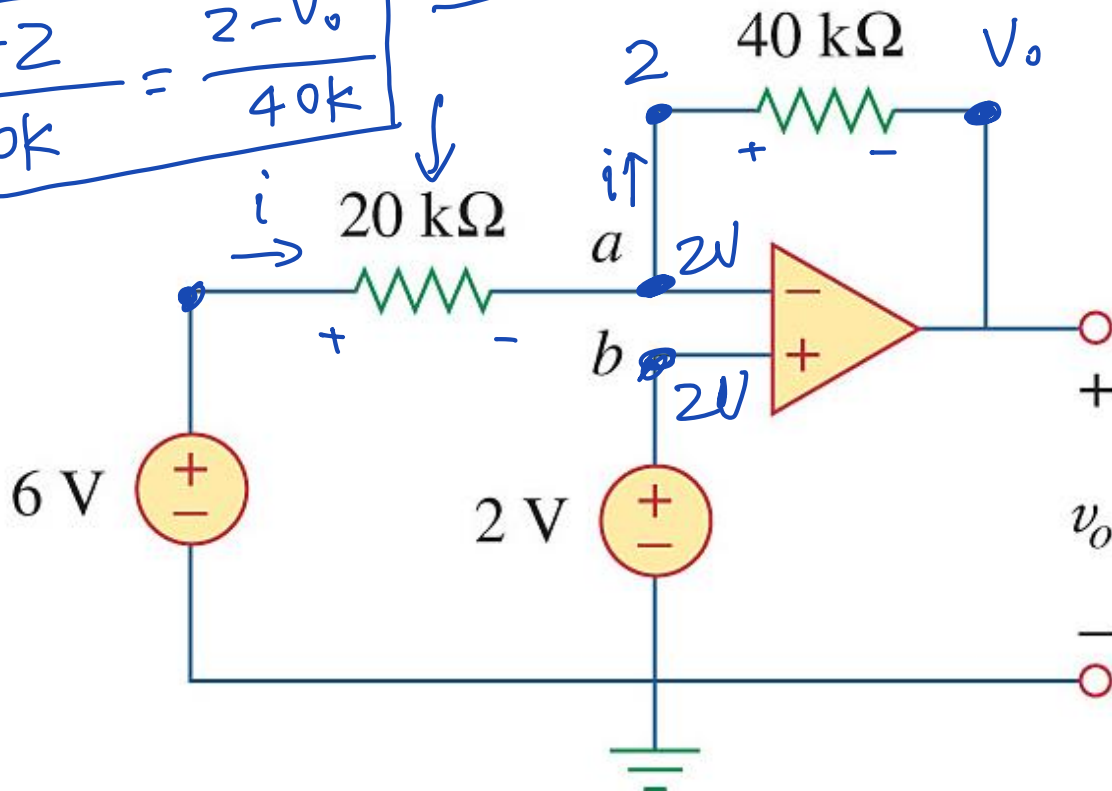
$$= - \frac{120 \text{ k}}{3 \text{ k}} \cdot 30 \times 10^{-3} \text{ V}$$

$$= -40 \times 30 \times 10^{-3} \text{ V} = \underline{\underline{-1.2 \text{ V.}}}$$

Example #2: Determine V_o

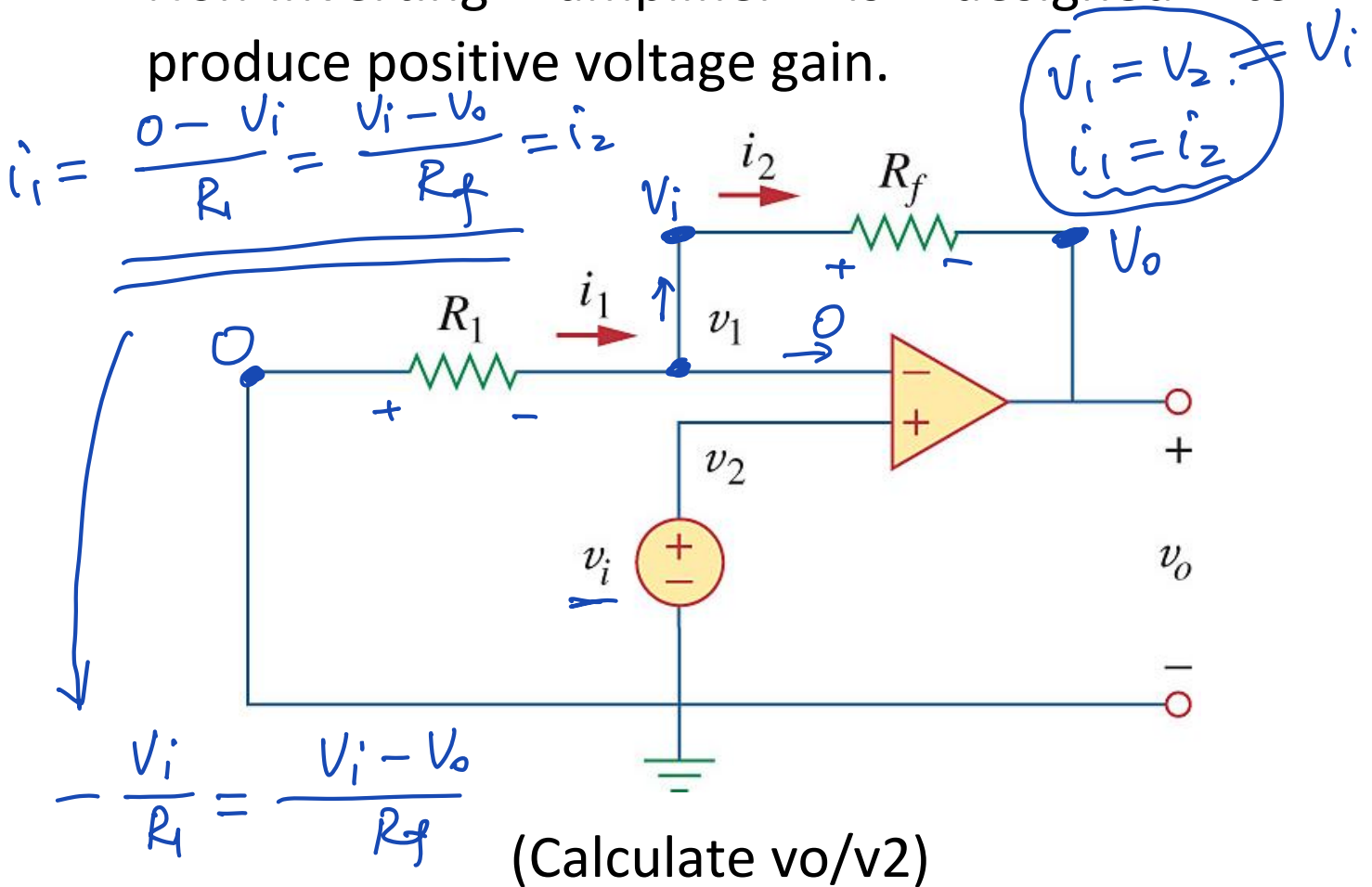
$$2(6-2) = 2 - V_o = 8 \Rightarrow V_o = -6V$$

$$i = \frac{6-2}{20k} = \frac{2-V_o}{40k}$$



Non-inverting amplifier

Non-inverting amplifier is designed to produce positive voltage gain.



$\times R_1 R_f$

$-R_f V_i = R_1 (V_i - V_o) = R_1 V_i - R_1 V_o$

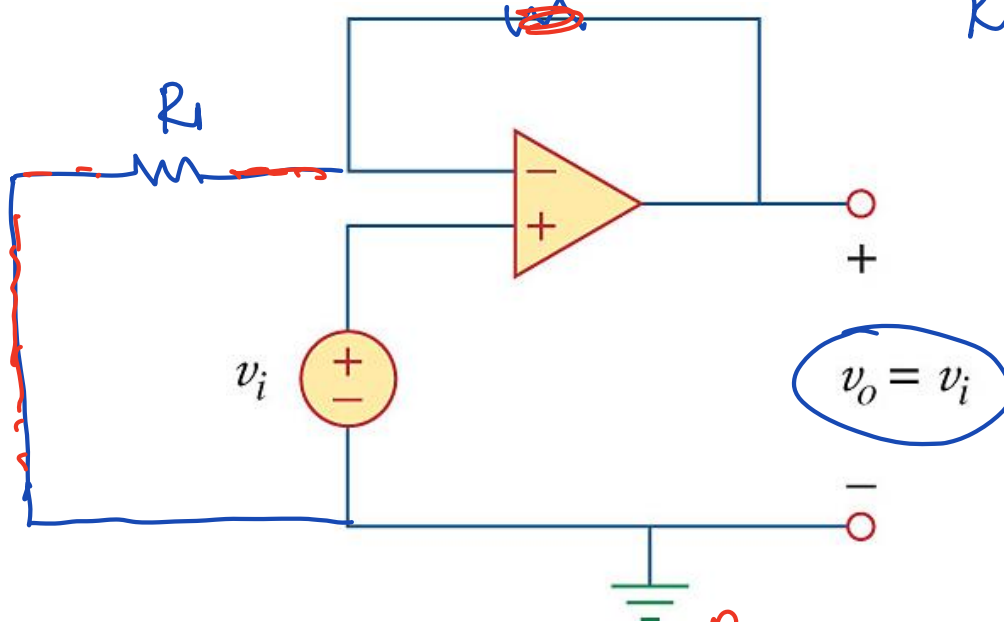
$R_1 V_o = R_1 V_i + R_f V_i = V_i (R_1 + R_f)$

$V_o = V_i \frac{R_1 + R_f}{R_1} = V_i \left(1 + \frac{R_f}{R_1} \right)$

Voltage follower:

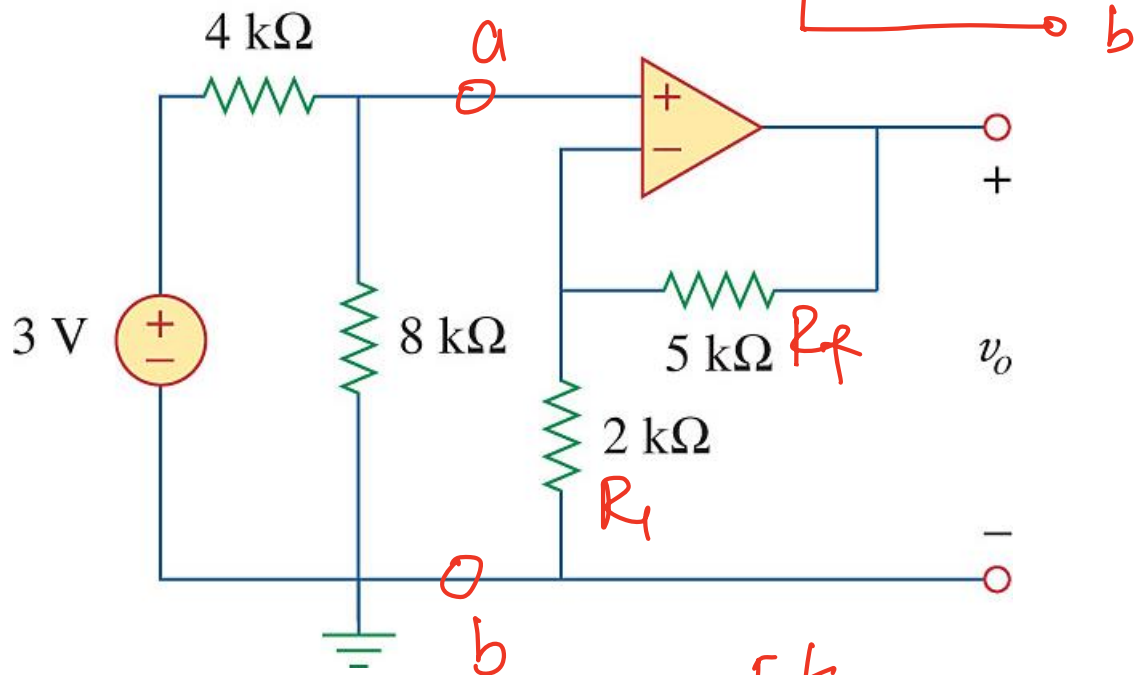
$$\rightarrow R_f = 0 \Omega.$$

$$R_i \rightarrow \infty$$

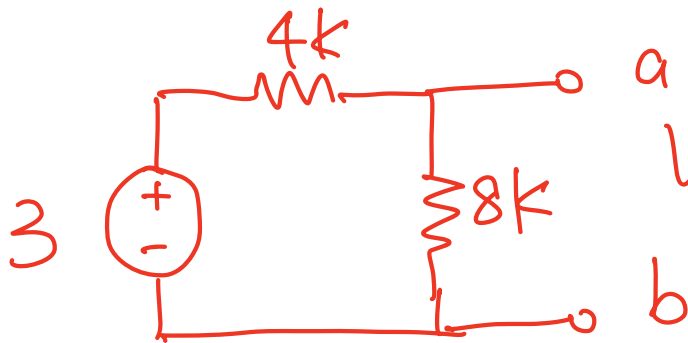


$$U_o = U_i \left(1 + \frac{R_f}{R_i} \right) = U_i \left(1 + \frac{0}{\infty} \right) = U_i$$

Example #3: Determine V_o .



$$V_o = V_{Th} \cdot \left(1 + \frac{5k}{2k}\right).$$



$$\Rightarrow V_{Th} = V_{ab}$$

$$= 3 \cdot \frac{8}{4+8}$$

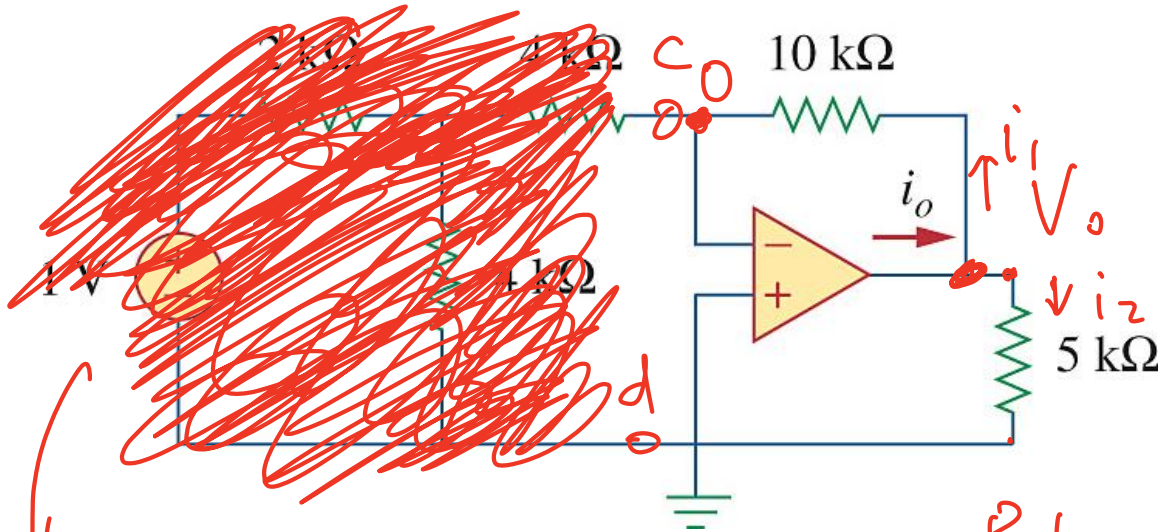
$$= 2V.$$

$$V_o = 2 \cdot (1 + 2.5) = 7V.$$

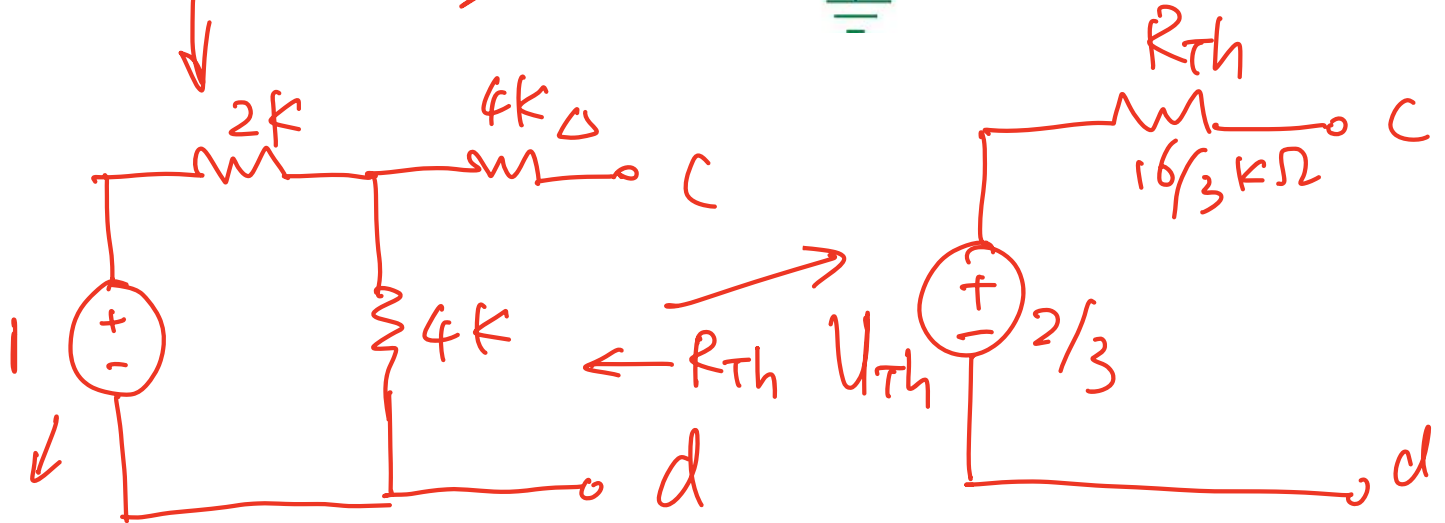
Example #4: Determine i_o

$$i_o = i_1 + i_2$$

$$= \frac{V_o - 0}{10k} + \frac{V_o - 0}{5k}$$



$$V_o = -1.25$$



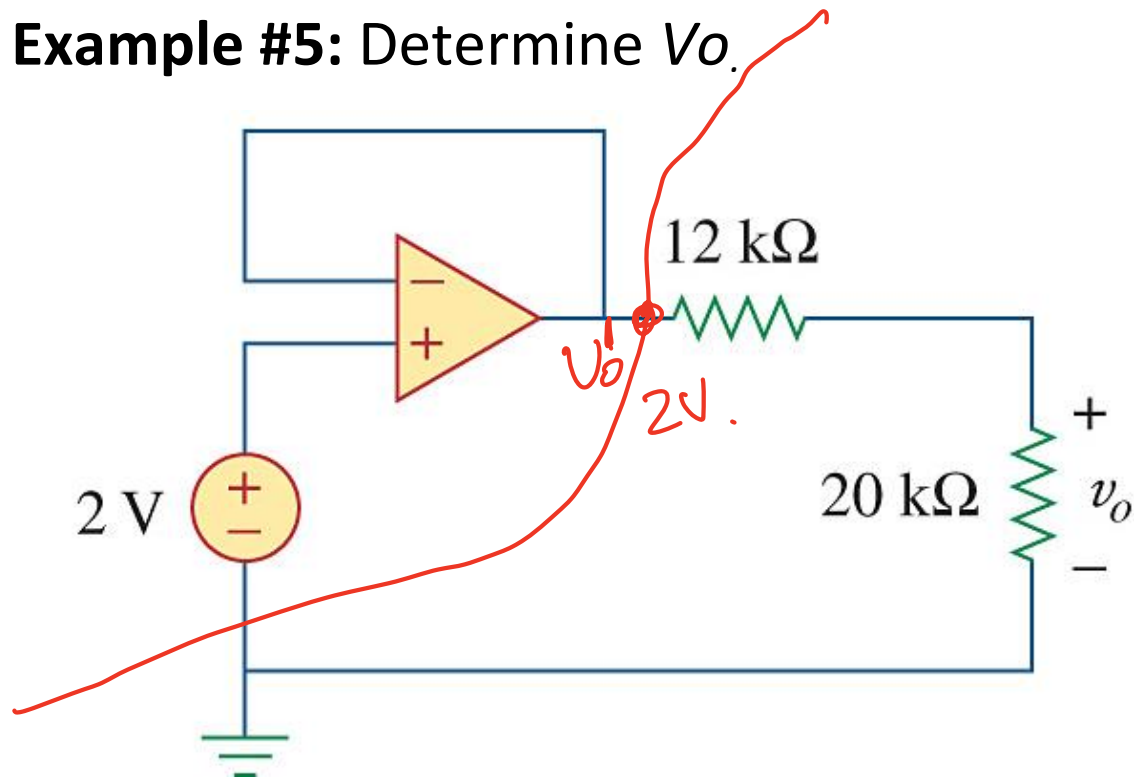
$$R_{Th} = 4 + \frac{2 \times 4}{2 + 4} = 4 + \frac{4}{3} = \frac{16}{3} k\Omega$$

$$V_{Th} = 1 \cdot \frac{4k}{2k + 4k} = \frac{2}{3} V$$

$$R_i = R_{Th}$$

$$V_o = -\frac{2}{3} \cdot \frac{R_f}{R_i} = -\frac{2}{3} \cdot \frac{10}{16/3} = -1.25 V$$

Example #5: Determine V_o .



$$V_o = V_o' \cdot \frac{20k}{12k + 20k}$$

↓
2V.